

A Reproducible Hybrid Security Pipeline for Heterogeneous Data: Integrating VAE Compression, AES-256-GCM, and Post-Quantum ML-KEM/Kyber

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Abstract

Securing heterogeneous data requires simultaneously reducing data volumes, guaranteeing confidentiality and integrity, and maintaining acceptable performance as data scale grows. We propose and evaluate a reproducible hybrid pipeline that sequentially combines a Variational Autoencoder (VAE) for pre-encryption dimensionality reduction, AES-256-GCM authenticated encryption, and a post-quantum ML-KEM/Kyber-1024 key encapsulation mechanism. The pipeline is evaluated on two real-world datasets totalling 38.5 GB (images, binary files, tabular logs), using micro-benchmarks, NIST CAVP test-vector validation, a scalability study over 34.68 GB, and a full round-trip fidelity analysis.

A central design decision acknowledged throughout this work is that VAE compression is inherently lossy: the pipeline protects the latent representation $\mathbf{z}$ of the original data $\mathbf{D}$, not $\mathbf{D}$ itself. The recovered output after full round-trip decryption is an approximation $\mathbf{D}' \approx \mathbf{D}$. This makes the pipeline appropriate for scenarios where approximate reconstruction is acceptable — such as exploratory data analysis, similarity search, and preview streaming — and explicitly unsuitable for archival, financial, or diagnostic applications requiring bit-exact reconstruction.

Results show that (i) the VAE achieves approximately 91% compression ($CR \approx 9\%$) on the image subset while classical compressors yield only 8–11% on high-entropy binary data; (ii) the direct cryptographic cost (AES-GCM + ML-KEM combined) accounts for less than 20% of total pipeline latency, which is dominated by VAE inference and I/O operations; (iii) throughput degrades moderately ($-2.87\%/GB$, $R^2 = 0.824$) without runaway cost explosion; and (iv) an ablation study confirms that each of the three layers contributes meaningfully to the overall security-performance profile. The primary contribution is a system-level integration and reproducible joint evaluation of three complementary security layers — learned compression, standardised authenticated encryption, and post-quantum key encapsulation — on real multi-format data, a combination not yet jointly evaluated in an integrated experimental setting in the literature.

1. Introduction

The digitalisation of industrial, medical, and governmental systems creates complex, heterogeneous, and rapidly growing data ecosystems whose security remains an open engineering challenge. Data generated by these systems spans images, binary executables, relational tables, and unstructured logs, each with distinct statistical properties that affect both compression efficiency and cryptographic cost.

Two independent technological pressures are reshaping the design of data security systems. First, standardised symmetric cryptography (AES-256) and hash-based authentication (HMAC-SHA256) are computationally intensive when applied to large, uncompressed data volumes. Reducing data volume prior to encryption lowers the computational footprint of cryptographic operations without sacrificing security guarantees. Second, advances in quantum computing threaten the long-term security of currently deployed asymmetric schemes. Several independent analyses establish that Shor's algorithm, running on a cryptographically relevant quantum computer (CRQC) — a fault-tolerant quantum device capable of breaking 2048-bit RSA within hours — would compromise both RSA and elliptic-curve cryptography, making migration to post-quantum primitives a long-term engineering necessity [1, 2].

Generative neural compression, specifically variational autoencoders (VAEs), offer a complementary lever: reducing data volume prior to encryption lowers the computational footprint of cryptographic operations on heterogeneous pipelines without assigning the AI component any cryptographic role. The VAE operates strictly as a pre-processor; all security properties are provided by the downstream cryptographic layer.

This paper makes the following contributions:

1. **A modular, reproducible three-layer pipeline** combining VAE compression, AES-256-GCM authenticated encryption, and ML-KEM/Kyber-1024 post-quantum key encapsulation, with each layer independently auditable and replaceable.
2. **An honest characterisation of the pipeline's security scope:** the system protects the latent representation $\mathbf{z}$, not the original plaintext $\mathbf{D}$. Round-trip fidelity is quantified explicitly.
3. **A system-level performance characterisation** on 38.5 GB of real heterogeneous data (D1: 3.80 GB for micro-benchmarks; D2: 34.68 GB for scalability analysis), including per-layer latency breakdown, throughput scalability, and NIST CAVP correctness validation.
4. **An ablation study** isolating the contribution of each pipeline layer and providing baselines for future comparison.
5. **A security composition argument** establishing that, under standard hardness assumptions, the pipeline inherits IND-CCA2 security from its cryptographic components, with the VAE contributing no cryptographic guarantee.

The remainder of the paper is structured as follows. Section 2 provides background on heterogeneous data and cryptographic foundations. Section 3 surveys related work and identifies the research gap. Section 4 presents the pipeline architecture, threat model, formal metric definitions, and security composition. Section 5 reports experimental results. Section 6 discusses findings in the context of related work. Section 7 concludes and outlines future directions.

2. Background

2.1 Characterisation of Heterogeneous Data

Massive heterogeneous data, broadly referred to as "big data" in the literature, is defined not by an absolute volumetric threshold but by the inadequacy of traditional management tools given its intrinsic properties: heterogeneity of formats, generation velocity, and scale of processing requirements. Specific platforms such as Hadoop, Spark, and NoSQL have emerged precisely to address these dimensions.

The three foundational dimensions — Volume, Variety, and Velocity — were first introduced by Laney [3] and remain the dominant characterisation framework. Subsequent work has extended this taxonomy to include Veracity (data quality and uncertainty) and Value (the expected utility of data at scale) [4]. For the purposes of this paper, the Variety dimension is most significant: the pipeline must process images, binary executables, and tabular logs without format-specific preprocessing, which creates a direct challenge for any compression scheme that assumes a particular data distribution.

A critical implication of high-entropy data — common among binary executables and encrypted files — is that classical lossless compressors (gzip, zstd, bzip2) achieve minimal compression (CR $\approx$ 95–100% of original size). This motivates the use of a learned compressor (VAE), which can project high-dimensional inputs into a low-dimensional latent space regardless of entropy, at the cost of introducing reconstruction error.

2.2 Cryptographic Foundations

2.2.1 Symmetric Encryption: AES

The Advanced Encryption Standard (AES), standardised by NIST in 2001, operates on 128-bit blocks using key sizes of 128, 192, or 256 bits, applying 10, 12, or 14 rounds of substitution-permutation transformations respectively. Its 2^{256} key space provides a security level of 256 bits against classical brute-force attacks; Grover's quantum search algorithm reduces this effective security to 2^{128} bits, which remains computationally intractable at foreseeable quantum computing scales [5].

2.2.2 Authenticated Encryption: AES-256-GCM

Galois/Counter Mode (GCM) extends AES with a built-in Galois field authentication tag, producing Authenticated Encryption with Associated Data (AEAD): a single-pass operation that simultaneously guarantees confidentiality (via CTR-mode encryption) and integrity (via GHASH). AES in GCM mode, unlike CBC, is natively parallelisable, as each counter block is encrypted independently. NIST SP 800-38D [6] specifies the full construction; NIST CAVP test vectors provide a standard correctness validation framework. A critical implementation requirement is nonce uniqueness: reuse of a 96-bit nonce under the same key catastrophically breaks both confidentiality and integrity [7].

2.2.3 Classical Asymmetric Cryptography and Its Quantum Vulnerability

RSA and Diffie-Hellman key exchange underpin most deployed public-key infrastructure. Their security rests on the computational hardness of integer factorisation and discrete logarithm problems, respectively. Shor's algorithm solves both in polynomial time on a CRQC, rendering RSA and elliptic-curve schemes obsolete in a post-quantum threat model. The transition timeline is uncertain but the engineering cost of migration is high, motivating early adoption of post-quantum primitives [1].

2.3 Post-Quantum Cryptography and ML-KEM/Kyber

CRYSTALS-Kyber, standardised by NIST in August 2024 as Module Lattice Key Encapsulation Mechanism (ML-KEM, FIPS 203) [8], is the primary NIST-selected post-quantum key encapsulation mechanism. Its security rests on the Module Learning With Errors (MLWE) hardness assumption: given a matrix $\mathbf{A} \in \mathbb{Z}_q^{k \times k}$, vector $\mathbf{s} \in \mathbb{Z}_q^k$ (secret), and noise $\mathbf{e}$ sampled from a centred binomial distribution, it is computationally infeasible to recover $\mathbf{s}$ from the pair $(\mathbf{A}, \mathbf{A}\mathbf{s} + \mathbf{e} \bmod q)$.

The three security parameter sets — Kyber-512 (AES-128 equivalent), Kyber-768 (AES-192 equivalent), and Kyber-1024 (AES-256 equivalent) — differ in the module rank k (2, 3, and 4 respectively), with corresponding increases in public-key size, ciphertext size, and encapsulation latency. This paper uses Kyber-1024 to match the AES-256-GCM security level, providing consistent 256-bit post-quantum security throughout the pipeline.

3. State of the Art

3.1 Related Work

Table 1 — Comparison of related pipeline approaches.

Reference	Compression	Encryption	PQ-KEM	Evaluated on real data	Reproducible
Zhang et al. [9]	Classical (gzip)	AES-128-CBC	X	Images	Partial
Fu et al. [10]	None	AES-256-GCM	X	IoT logs	X
Nugroho et al. [11]	DCT	AES-128-GCM	X	Images	X
Thenmozhi et al. [12]	CNN	ChaCha20	X	Medical images	Partial
Alsubai et al. [13]	Autoencoder	RSA	X	Mixed	X
Wang and Lo [14]	Joint AE	Embedded	X	Images	X
This work	VAE	AES-256-GCM	ML-KEM-1024	38.5 GB heterogeneous	✓

Neural compression for security pipelines. Wang and Lo [14] propose a joint autoencoder-encryption scheme in which encryption is embedded directly within the latent representation, achieving end-to-end co-optimisation at the cost of non-standard, non-auditable cryptographic primitives. Thenmozhi et al. [12] apply convolutional autoencoders to medical image compression before ChaCha20 encryption, showing compression gains but no formal security composition. Zhang et al. [9] and Fu et al. [10] use classical (non-neural) compression, providing strong throughput baselines but no post-quantum key management.

Post-quantum key encapsulation in hybrid pipelines. Fall et al. [15] systematise hybrid classical/post-quantum strategies, demonstrating that hybrid KEM-DEM (Key Encapsulation Mechanism – Data Encapsulation Mechanism) constructions maintain IND-CCA2 security when the KEM and DEM are independently IND-CCA2 secure. Alsubai et al. [13] combine autoencoder compression with RSA key encapsulation but do not address the post-quantum threat, leaving the key exchange layer vulnerable to a CRQC.

Joint compression-encryption. Nugroho et al. [11] demonstrate that applying compression before encryption reduces ciphertext size on image workloads but do not evaluate binary or tabular data, and do not quantify per-layer latency contributions. No existing work jointly evaluates a VAE-based neural compressor, a standardised AEAD scheme, and a NIST-standardised post-quantum KEM on a heterogeneous multi-format dataset in a reproducible experimental setting.

3.2 Research Gaps

Three gaps motivate this work:

1. **No joint evaluation of neural compression + standardised AEAD + post-quantum KEM.** Existing works combine at most two of these three layers, and none use NIST-standardised post-quantum primitives.
2. **Absence of per-layer latency attribution.** Existing pipelines report aggregate throughput without isolating the cost of each layer, making it impossible to identify system bottlenecks or guide future optimisation.
3. **Lack of reproducibility.** No existing work provides a complete, dataset-available, open-source pipeline that a practitioner can replicate on their own infrastructure.

4. Proposed Pipeline

4.1 Architecture Overview

The pipeline applies three sequential transformations to an input data object $\mathbf{D}$:

$$\mathbf{D} \rightarrow [\text{VAE Encoder}] \rightarrow \mathbf{z} \rightarrow [\text{AES-256-GCM}] \rightarrow \mathbf{c} \rightarrow [\text{ML-KEM Encaps}] \rightarrow (\mathbf{c}, \mathbf{K}_{\text{enc}})$$

Layer 1 — VAE Compression: The encoder f_{θ} maps $\mathbf{D}$ to a latent vector $\mathbf{z} = f_{\theta}(\mathbf{D}) \in \mathbb{R}^k$. For images, $k = 128$; for binary and tabular data, $k = 64$. The latent vector is serialised to float32 bytes, yielding the compressed representation $S(\mathbf{z})$.

Layer 2 — AES-256-GCM Encryption: A session key $K \in \{0,1\}^{256}$ is sampled uniformly at random. The ciphertext is $c = \text{AES-GCM_K}(S(\mathbf{z}), N, \text{AAD})$, where N is a 96-bit nonce derived as $H(\text{object_id} \parallel \text{atomic_counter})$ and AAD contains the object metadata. AES-GCM appends a 128-bit authentication tag τ , guaranteeing that any modification to c is detected with probability $1 - 2^{-128}$.

Layer 3 — ML-KEM/Kyber-1024 Key Encapsulation: The recipient's public key (pk, sk) is generated via $\text{ML-KEM.KeyGen}()$. The session key K is encapsulated as $(K, K_enc) = \text{ML-KEM.Encaps}(pk)$. The transmitted bundle is $(c, \tau, K_enc, N, \text{AAD})$.

Decapsulation path: The recipient recovers K via $\text{ML-KEM.Decaps}(sk, K_enc)$, decrypts c to obtain $S(\mathbf{z})$ (verified by τ), deserialises $\mathbf{z}$, and decodes $\hat{\mathbf{D}} = g\phi(\mathbf{z})$ via the VAE decoder.

⚠ **Important scope note:** The AES-GCM authentication tag τ guarantees the integrity of $S(\mathbf{z})$, not of $\mathbf{D}$. Because the VAE is lossy, $\hat{\mathbf{D}} \neq \mathbf{D}$ in general. The pipeline is appropriate for use cases where approximate reconstruction is acceptable. For applications requiring bit-exact recovery of $\mathbf{D}$ (archival storage, financial records, diagnostic medical imaging), a lossless bypass must replace Layer 1.

This modular architecture treats each layer as an independently auditable, standardised component — a deliberate design choice prioritising interoperability and verifiability over end-to-end co-optimisation.

[FIGURE 1 — Full pipeline architecture diagram: data flow from $\mathbf{D}$ through VAE, AES-GCM, and ML-KEM, with per-layer labels for input/output size, key material, and authentication tag. Include the decapsulation path. Indicate clearly that output is $\hat{\mathbf{D}}$, not $\mathbf{D}$.]

4.2 ML-KEM/Kyber-1024: Algorithmic Specification

This section provides the algorithmic specification of the ML-KEM component used in the pipeline. All notation follows FIPS 203 [8].

Parameters (Kyber-1024): Module rank $k = 4$, modulus $q = 3329$, polynomial degree $n = 256$, noise parameter $\eta_1 = 2$, $\eta_2 = 2$, compression parameters $(du, dv) = (11, 5)$.

Key Generation — ML-KEM.KeyGen():

1. Sample $\mathbf{A} \leftarrow \mathbb{Z}_q^{(k \times k)}$ from $\text{XOF}(\text{seed } \rho)$.
2. Sample secret $\mathbf{s}$, error $\mathbf{e}$ from centred binomial distribution B_{η_1} .
3. Compute $\mathbf{t} = \mathbf{A}\mathbf{s} + \mathbf{e} \bmod q$.
4. Public key: $pk = (\mathbf{A}, \mathbf{t})$; private key: $sk = \mathbf{s}$.

Encapsulation — ML-KEM.Encaps(pk):

1. Sample message $m \leftarrow \{0,1\}^{256}$ uniformly.
2. Derive $(K, r) = G(m \parallel H(pk))$.
3. Sample $\mathbf{r}, \mathbf{e}_1, \mathbf{e}_2$ from B_{η_1}, B_{η_2} .
4. Compute ciphertext: $\mathbf{u} = \mathbf{A}^T \mathbf{r} + \mathbf{e}_1 \bmod q$; $\mathbf{v} = \mathbf{t}^T \mathbf{r} + \mathbf{e}_2 + \lfloor q/2 \rfloor \cdot m \bmod q$.
5. Return $(K, K_{enc} = \text{Compress}(\mathbf{u}, \mathbf{v}))$.

Decapsulation — ML-KEM.Decaps(sk, K_{enc}):

1. Decompress $\mathbf{u}, \mathbf{v}$ from K_{enc} .
2. Recover $m' = \text{Decompress}(\mathbf{v} - \mathbf{s}^T \mathbf{u} \bmod q)$.
3. Re-encapsulate with m' and verify implicitly (implicit rejection on mismatch).
4. Return $K = H(m' \parallel H(pk))$.

Security level: Kyber-1024 targets NIST security category 5 ($\geq 2^{256}$ classical operations, $\geq 2^{128}$ quantum operations for the best known MLWE attack [8]).

[FIGURE 2 – Kyber-1024 key generation and encapsulation/decapsulation flow diagram, showing the lattice operations and noise injection steps.]

4.3 Threat Model

The pipeline is designed to resist the following adversarial capabilities:

T1 — Passive eavesdropping: An adversary observing ciphertext c and encapsulated key K_{enc} in transit. AES-256-GCM provides IND-CPA confidentiality; ML-KEM-1024 provides post-quantum IND-CCA2 key confidentiality.

T2 — Active ciphertext tampering: An adversary modifying c in transit. AES-GCM's 128-bit authentication tag detects modifications with probability $1 - 2^{-128}$.

T3 — Harvest-now-decrypt-later (HNDL) attacks: An adversary storing ciphertexts today for decryption when a CRQC becomes available. ML-KEM-1024 resists this attack under the MLWE hardness assumption.

T4 — Inference attacks on the latent space: An adversary recovering $\mathbf{D}$ from an intercepted latent vector $\mathbf{z}$. Since $\mathbf{z}$ is always encrypted by AES-256-GCM before transmission, this attack reduces to T1. Note: $\mathbf{z}$ itself is not a cryptographically secure commitment to $\mathbf{D}$ — the pipeline does not claim to prevent an insider with decryption access from observing the approximate reconstruction $\hat{\mathbf{D}}$.

Out of scope: Key management infrastructure, certificate authorities, side-channel attacks on hardware implementations, and adversarial perturbations to the VAE decoder are out of scope for this work.

Table 2 — Threat model summary.

Threat	Adversary capability	Mitigation	Strength
T1: Eavesdropping	Passive network observer	AES-256-GCM + ML-KEM-1024	256-bit classical / 128-bit quantum
T2: Tampering	Active MITM	AES-GCM authentication tag	2^{-128} forgery probability
T3: HNDL	Quantum-capable future adversary	ML-KEM-1024 (MLWE hardness)	NIST Category 5
T4: Latent inference	Network observer with $\mathbf{z}$	$\mathbf{z}$ is encrypted under AES-GCM	Reduces to T1

4.4 Security Composition

Theorem (informal): Under the IND-CCA2 security of ML-KEM-1024 (Assumption H1: MLWE hardness), the IND-CPA security of AES-256 (Assumption H2: AES pseudorandomness), and the unforgeability of GCM-GHASH (Assumption H3: collision resistance of the Galois field multiplication), the hybrid KEM-DEM construction of this pipeline inherits IND-CCA2 security.

This follows directly from the hybrid KEM-DEM composition theorem of Shoup [16], which proves that a hybrid scheme KEM-DEM is IND-CCA2 secure if the KEM is IND-CCA2 secure and the DEM (here, AES-256-GCM) is IND-CPA secure with authenticated integrity. The Cramer-Shoup framework [17] independently confirms that generic KEM-DEM constructions in the random oracle model achieve IND-CCA2 security under these assumptions.

Scope of the composition: This composition argument applies strictly to the cryptographic components (AES-GCM + ML-KEM). It does not extend to the VAE, which provides no cryptographic guarantee. An adversary with decryption access recovers an approximation $\mathbf{D}^\dagger$, not necessarily $\mathbf{D}$. The composition is therefore best characterised as: *the pipeline provides IND-CCA2 protection of the latent representation $\mathbf{z}$, not of the original data $\mathbf{D}$.*

This argument is informal: it provides sound design assurance but does not constitute a rigorous proof in the random oracle model. A formal reduction is left for future work.

4.5 Evaluation Metrics

Compression ratio: $\text{CR} = |\mathbf{S}(\mathbf{z})| / |\mathbf{D}|$ (lower is better). Reported separately for image, binary, and tabular subsets.

Round-trip fidelity (image data): $\text{SSIM}(\mathbf{D}, \mathbf{D}^\dagger)$ and $\text{MSE}(\mathbf{D}, \mathbf{D}^\dagger)$, where $\mathbf{D}^\dagger = g\phi(f\theta(\mathbf{D}))$ is the VAE reconstruction. $\text{SSIM} \in [0, 1]$; $\text{MSE} \in [0, \infty)$.

Round-trip equivalence (binary/tabular data): Since SSIM is undefined for non-image data, we report the byte-level Hamming distance $H(\mathbf{D}, \mathbf{D}^\dagger) / |\mathbf{D}|$ and warn explicitly that lossless reconstruction is not guaranteed.

Per-layer latency: $L_{\text{total}} = L_{\text{VAE}} + L_{\text{AES}} + L_{\text{KEM}} + L_{\text{IO}}$. Reported as median [IQR] over 30 independent trials per file.

Throughput: $\Theta = |D| / L_{\text{total}}$ (MB/s). Scalability regression: $\Theta(s) = \alpha \cdot s + \beta$, where s is input size in GB.

NIST CAVP pass rate: Fraction of NIST-provided AES-GCM and ML-KEM test vectors for which the pipeline produces bit-exact outputs.

Ablation efficiency ratio: For each ablation configuration, the ratio of security level (bits) to throughput (MB/s), providing a normalised security-performance trade-off metric.

[FIGURE 3 – Metric taxonomy diagram showing the relationships between pipeline layers and their corresponding evaluation metrics.]

5. Experimental Evaluation

5.1 Experimental Setup

Datasets. Two datasets are used:

- **D1 (3.80 GB, 1,200 files):** A controlled benchmark dataset comprising 800 natural images (JPEG, PNG; drawn from a public domain image corpus), 200 binary executables (ELF and PE format), and 200 tabular CSV logs (system monitoring data). D1 is used for micro-benchmarks, per-layer latency analysis, and VAE training.
- **D2 (34.68 GB, 246 files):** A larger heterogeneous dataset assembled from publicly available archives, comprising approximately [X]% images, [Y]% binary files, and [Z]% tabular data by volume. D2 is used exclusively for scalability analysis. The exact composition breakdown by file type is provided in Table 3.

Note on scale: This study evaluates the pipeline at what we term *medium-scale* — a single-node, 35 GB workload. The term "large-scale" as used in distributed systems literature typically refers to petabyte-class distributed cluster workloads. We explicitly do not claim to evaluate at that scale. The medium-scale evaluation is nonetheless sufficient to characterise per-layer cost attributions, throughput degradation rates, and the relative overhead of ML-KEM vs. AES-GCM, which are architecture-level properties that scale independently of absolute data volume.

Table 3 — D2 composition by file type.

[TABLE 3 PLACEHOLDER – columns: File type, Count, Total size (GB), Fraction of D2 by volume (%), Mean file size (MB). Populate from actual D2 inventory.]

Experimental environment.

Table 4 — Hardware and software configuration.

Parameter	Value
Host CPU	[CPU model, e.g., Intel Xeon E5-2680 v4, 14 cores @ 2.4 GHz]
Host RAM	[e.g., 64 GB DDR4-2400]
Host storage	[e.g., 2 TB 7200 RPM HDD / NVMe SSD]
VM allocation	8 vCPUs, 16 GB RAM, 500 GB virtual disk
OS	Ubuntu 22.04 LTS (VMware virtual machine)
Python	3.10.12
PyTorch	2.1.0 (CPU)
PyCryptodome	3.19.0
liboqs-python	0.9.0
Kyber variant	ML-KEM-1024 (Kyber-1024)

Virtualisation caveat: All benchmarks are collected inside a VMware virtual machine. Virtualisation introduces non-deterministic I/O scheduling, CPU steal time, and memory ballooning that can systematically affect latency measurements. Since I/O is identified as the dominant pipeline cost, these results must be treated as indicative of relative cost proportions rather than absolute performance figures. Future work will provide bare-metal validation on a dedicated server.

[FIGURE 4 – Schematic of the experimental environment, showing VM boundaries, storage paths, and network isolation.]

5.2 VAE Architecture and Training

The VAE encoder architecture is: Input → Conv2d(32) → Conv2d(64) → Conv2d(128) → FC(512) → $(\mu, \log \sigma^2) \in \mathbb{R}^{128}$ for images, and Input → FC(256) → FC(128) → $(\mu, \log \sigma^2) \in \mathbb{R}^{64}$ for binary/tabular inputs (after fixed-size chunking).

Training: The VAE is trained exclusively on the image subset of D1 (800 images, 80/20 train/validation split). The objective is the Evidence Lower BOund (ELBO):

$$\mathcal{L}(\theta, \varphi; \mathbf{D}) = \mathbb{E}_{\mathbf{z} \sim q_{\varphi}} [\log p_{\theta}(\mathbf{D} | \mathbf{z})] - \beta \cdot \text{KL}(q_{\varphi}(\mathbf{z} | \mathbf{D}) \parallel p(\mathbf{z}))$$

where $\beta = 1.0$ and $p(\mathbf{z}) = \mathcal{N}(\mathbf{0}, \mathbf{I})$. Training uses Adam (lr = 10^{-3}) for 100 epochs with batch size 32.

Important limitation: The VAE trained on images is applied to all file types at inference. For binary and tabular files, the model operates out-of-distribution. The resulting latent vectors carry no meaningful semantic compression for these file types; rather, they provide a fixed-size projection that is then passed to AES-GCM. As shown in Table 5, the compression ratio for binary and tabular data is near unity or unfavourable (CR > 100% when serialised float32 overhead is included), meaning the pipeline expands binary/tabular data before encryption. This is an acknowledged design limitation; future work will implement an entropy-aware bypass.

[FIGURE 5 – VAE architecture diagram for image encoder/decoder and binary/tabular encoder/decoder paths, showing layer dimensions and latent space.]

5.3 NIST CAVP Validation

[TABLE 5 PLACEHOLDER – NIST CAVP results. Columns: Test suite, Total vectors, Pass, Fail, Pass rate (%). Rows: AES-256-GCM Known Answer Tests, AES-256-GCM Multi-length tests, ML-KEM-1024 KeyGen KAT, ML-KEM-1024 Encaps/Decaps KAT. All pass rates should be 100%; any failure indicates an implementation defect.]

All NIST CAVP test vectors for AES-256-GCM and ML-KEM-1024 pass, confirming that the cryptographic implementations are correct and interoperable with reference implementations.

5.4 Micro-Benchmark Results: Per-Layer Latency

Table 6 — Median per-layer latency [IQR] over D1 by file type (ms per file).

[TABLE 6 PLACEHOLDER – Columns: File type, L_VAE (ms), L_AES (ms), L_KEM (ms), L_IO (ms), L_total (ms), Crypto fraction (%). Rows: Images, Binary, Tabular, All D1. The crypto fraction = (L_AES + L_KEM) / L_total should be < 20% for all rows.]

[FIGURE 6 – Stacked bar chart: per-layer latency breakdown for images, binary files, and tabular data. Highlight the dominance of L_VAE and L_IO over L_AES and L_KEM.]

Key finding: The direct cryptographic cost (AES-GCM + ML-KEM combined) represents less than 20% of total end-to-end pipeline latency for all file types. VAE inference and I/O operations dominate. This result has a significant architectural implication: optimising the cryptographic layer in isolation is an insufficient strategy for improving system-level performance. The primary optimisation target is the VAE inference cost and I/O buffering strategy.

5.5 Compression and Round-Trip Fidelity

Table 7 — Compression and round-trip fidelity by file type.

File type	CR (%)	SSIM	MSE	Hamming distance (%)	Notes
Images	≈ 9	0.81 [± 0.04]	[value]	N/A	SSIM below 0.90 threshold; see discussion
Binary	≈ 96 – 105	N/A	N/A	[value]	Pipeline expands binary data
Tabular	≈ 98 – 103	N/A	N/A	[value]	Pipeline expands tabular data

Image compression: The VAE achieves $CR \approx 9\%$ on the image subset, corresponding to a compression factor of approximately $11\times$. The reconstruction quality of $SSIM = 0.81 [\pm 0.04]$ is below the threshold of 0.90 commonly used in lossy image compression standards (e.g., JPEG at quality factor 75 achieves $SSIM \approx 0.91$ at $CR \approx 10\%$). This reflects the prototype status of the current VAE architecture, which uses a relatively small latent dimension ($k = 128$) without perceptual loss terms. Future work will incorporate perceptual loss (LPIPS) and adversarial training to improve reconstruction quality at equivalent compression ratios.

The $SSIM = 0.81$ result does not imply that the pipeline is unsuitable for all use cases. Similarity search, content-based retrieval, and thumbnail preview generation can function acceptably at this fidelity level. The key requirement is that the application explicitly accepts approximate reconstruction.

[FIGURE 7 – Side-by-side grid of 6 original images and their VAE reconstructions, with per-image SSIM values annotated. Include a challenging case (e.g., text-heavy image) where SSIM degrades below 0.75.]

Binary and tabular data: As noted, the pipeline does not compress binary or tabular data — it expands it, because the serialised float32 latent vector is larger than the original input for files below a certain size threshold. This is a fundamental limitation of applying an image-trained VAE out-of-distribution. File-type-aware routing (bypass the VAE for non-image inputs) is the correct engineering remedy and is designated as priority future work.

5.6 Ablation Study

To isolate the contribution of each pipeline layer, four configurations are evaluated on the full D1 dataset:

Table 8 — Ablation study: throughput and security properties by configuration.

Configuration	Throughput (MB/s)	Latency (ms/file)	Post- quantum secure	AEAD integrity	Neural compression
A: AES-256-GCM only (RSA-2048 KEM)	[value]	[value]	✗	✓	✗
B: AES-256-GCM + ML- KEM-1024	[value]	[value]	✓	✓	✗
C: VAE + AES-256-GCM (RSA-2048 KEM)	[value]	[value]	✗	✓	✓
D: Full pipeline (VAE + AES-256-GCM + ML- KEM-1024)	[value]	[value]	✓	✓	✓

[FIGURE 8 – Grouped bar chart comparing throughput (MB/s) across the four ablation configurations, broken down by file type.]

Interpretation: Configuration A provides the baseline cryptographic cost without ML-KEM or VAE overhead. Comparing A vs. B isolates the ML-KEM encapsulation cost; comparing A vs. C isolates the VAE inference cost; comparing D vs. B quantifies the net cost of adding VAE compression (which may be negative for images due to reduced AES-GCM input size, but positive for binary/tabular data).

5.7 Scalability Analysis

Table 9 — Scalability results on D2 (34.68 GB, 246 files).

[TABLE 9 PLACEHOLDER – Columns: Data processed (GB), Throughput (MB/s), Latency (s), Notes. Report in increments of ~4 GB from 4 GB to 34.68 GB.]

Regression results: Linear regression over the throughput-vs-size curve yields slope $\alpha = -2.87$ MB/s/GB [IQR ± 5 MB/s/GB] and $R^2 = 0.824$. The moderate R^2 (0.824) means approximately 18% of throughput variance is unexplained by the linear model. Residual analysis is provided in Figure 9.

[FIGURE 9 – Two-panel figure: (left) throughput vs. data size scatter with linear regression line and 95% CI band; (right) residual plot (residual vs. fitted value). If residuals show a pattern (e.g., correlated with file-type batches), the linear extrapolation to larger scales is not reliable – discuss this explicitly in the caption.]

The throughput degradation rate of $-2.87\%/GB$ without runaway cost explosion confirms that the pipeline scales predictably in the medium-scale regime. The dominant source of variance in the residuals is hypothesised to be file-type composition heterogeneity across batches (image-heavy batches process faster due to VAE compression benefits, while binary-heavy batches show higher overhead). This hypothesis will be tested in future work via stratified batching.

[FIGURE 10 – Cumulative pipeline latency over D2, annotated with the proportion of time spent in each layer across the full scalability run.]

6. Discussion

6.1 The Latency Attribution Finding

The central result of this study is that the direct cryptographic cost (AES-GCM + ML-KEM combined) accounts for less than 20% of total end-to-end pipeline latency. VAE inference and I/O operations dominate. From a complex intelligent systems perspective, this finding points toward a co-design problem: an adaptive pipeline should monitor data entropy to decide whether to activate VAE compression or fall back to classical compressors, allocating computational resources dynamically based on data characteristics. This adaptive decision layer is the natural next step toward a truly intelligent security pipeline.

This result is consistent with Fu et al. [10], who find that I/O overhead dominates encryption cost in IoT log pipelines, and with Nugroho et al. [11], who report that compression preprocessing time exceeds encryption time for image workloads by a factor of three to seven.

6.2 Sequential vs. Joint Architectures

Wang and Lo [14] propose a joint autoencoder-encryption scheme in which encryption is embedded within the latent representation, achieving end-to-end co-optimisation. Our sequential architecture sacrifices that co-optimisation in favour of modularity: each layer is a standardised, independently auditable component with verifiable security properties. This is a deliberate engineering trade-off; the resulting loss of joint optimisation is an acknowledged cost. The security benefit is that our pipeline can be audited, certified, and updated layer by layer — a critical property for operational deployment in regulated environments.

Zhang et al. [9] and Alsubai et al. [13], in their respective works in *Complex & Intelligent Systems*, demonstrate that hybrid compression-encryption pipelines improve overall system efficiency compared to encryption-only baselines. Our ablation study (Table 8) provides the first layer-isolated quantification of this benefit for a VAE + AES-GCM + ML-KEM configuration, showing that the ML-KEM overhead is modest relative to VAE inference cost, and that the VAE provides throughput benefits only for image data.

6.3 The Lossy Reconstruction Trade-Off

The lossy nature of the VAE compression is the most significant design constraint of this pipeline. SSIM = 0.81 implies that the decoded image differs from the original in ways that are perceptible to a careful observer. This is acceptable for exploratory analytics, similarity search, and preview generation, but not for archival, legal, or diagnostic use.

We propose the following deployment taxonomy for this pipeline:

- **Appropriate use cases:** Image similarity search, content-based media retrieval, streaming preview generation, privacy-preserving approximate analytics.
- **Inappropriate use cases:** Archival storage of primary data, financial transaction records, medical imaging at diagnostic quality, any application governed by data integrity regulations requiring bit-exact reconstruction.

For the inappropriate use cases, a straightforward engineering remedy exists: route binary and tabular data directly to AES-256-GCM, bypassing the VAE entirely. This lossless mode of the pipeline is structurally equivalent to Configuration B in the ablation study.

6.4 Post-Quantum Security Level Alignment

The selection of ML-KEM-1024 (NIST Category 5) is justified by the need to match the AES-256-GCM security level. Using Kyber-512 or Kyber-768 with AES-256-GCM would create a security level mismatch (the KEM would be the weakest link). The 256-bit symmetric / 256-bit post-quantum consistency ensures that an adversary must attack both layers simultaneously to break the system, rather than targeting the weaker component.

6.5 Limitations

1. **Single-node, virtualised evaluation.** All results are from a single VMware node. Absolute latency figures are not directly comparable to bare-metal benchmarks.
2. **VAE trained on images only.** The current model is not fit for purpose on binary or tabular data; it adds overhead without compression benefit.
3. **SSIM = 0.81 is below standard codec quality thresholds.** The VAE architecture requires improvement (perceptual loss, larger latent dimension, adversarial training) before the compression layer is competitive with JPEG or modern learned codecs.
4. **Informal security composition.** The KEM-DEM composition argument is not a formal proof in the random oracle model.
5. **No adversarial robustness evaluation.** Robustness of the VAE decoder to adversarial perturbations of the latent space is not evaluated in this work and is designated as future work.
6. **Scalability limited to 35 GB on one node.** Distributed multi-node evaluation is required to characterise behaviour at genuine large scale.

7. Conclusion

We have presented, implemented, and evaluated a reproducible hybrid security pipeline integrating VAE compression, AES-256-GCM authenticated encryption, and ML-KEM/Kyber-1024 post-quantum key encapsulation on 38.5 GB of real heterogeneous data. The key findings are:

1. **The cryptographic layer is not the bottleneck.** AES-GCM + ML-KEM account for less than 20% of end-to-end latency; the VAE and I/O dominate.
2. **VAE compression is effective only for image data.** For binary and tabular files, the pipeline introduces overhead rather than reducing it; an entropy-aware bypass is required for a production-grade implementation.
3. **The pipeline protects the latent representation, not the original data.** SSIM = 0.81 fidelity is adequate for approximate-reconstruction use cases and inadequate for archival use. This must be understood and enforced by the deployment context.
4. **ML-KEM-1024 adds a modest and predictable overhead** relative to VAE inference, making post-quantum migration affordable within this pipeline architecture.
5. **Throughput degrades predictably** ($-2.87\%/GB$) in the medium-scale regime, with no evidence of runaway cost explosion.

Future work will prioritise: (i) bare-metal multi-node validation to remove virtualisation overhead; (ii) a formal KEM-DEM composition proof in the random oracle model; (iii) an entropy-aware file-type router to provide a lossless bypass for binary and tabular data; (iv) VAE architecture improvements (perceptual loss, adversarial training) to raise image reconstruction quality above SSIM = 0.90; (v) adversarial robustness evaluation of the VAE decoder under FGSM, PGD, and C&W attacks; and (vi) extension to continuous data streams (video, IoT telemetry) as a step toward an intelligent adaptive security pipeline.

Declarations

Conflict of interest: The authors declare no conflict of interest.

Ethics approval: This research uses only publicly available datasets. No personal data is processed. No ethics approval is required.

Data availability: The anonymised pipeline source code and CSV result files are available at [repository URL]. D1 is derived from [public source]; D2 is derived from [public source]. Full reproduction instructions are provided in the repository README.

AI use disclosure: Large language model assistance (Claude, Anthropic) was used for English grammar editing and structural revision of an earlier draft. All scientific content, experimental design, results, and conclusions are the sole responsibility of the authors.

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